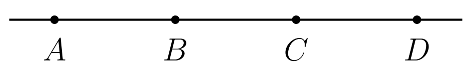
Completing Short Algebraic and Segment Proofs

Justifying each step of an algebraic solution, proving segment relationships with the Segment Addition Postulate, and using the definitions of midpoint and congruent segments

Every line of a proof needs a reason. A statement is what you know; a reason is the definition, postulate, property or theorem that lets you write it. The reasons you will need on this sheet:

- **Properties of equality** — Addition, Subtraction, Multiplication, Division, Substitution, and the Distributive Property.
- **Segment Addition Postulate** — if B is between A and C, then $AB + BC = AC$.
- **Definition of midpoint** — if M is the midpoint of $\overline{AB}$, then $\overline{AM} \cong \overline{MB}$.
- **Definition of congruent segments** — $\overline{AB} \cong \overline{CD}$ exactly when $AB = CD$.

All lengths on this sheet are in centimetres. Write the requested value on the answer line, and fill every blank cell of a proof table.



How points are labelled here: A, B, C, D lie on one line in that order, and AB means the length of the segment from A to B. No values shown — use the lengths given in each problem.

1. **Given:** $3x + 7 = 25$. **Prove:** $x = 6$.

Complete the Reasons column.

Statements	**Reasons**
1. $3x + 7 = 25$	1. Given
2. $3x = 18$	2.
3. $x = 6$	3.

Answer: ____________________

2. B is between A and C, with $AB = 2x + 5$, $BC = x + 3$, and $AC = 26$ centimetres.
(a) Name the postulate that lets you write $AB + BC = AC$.
(b) Write that equation, solve it, and give $x =$ ________
(c) Find the length AB.

Answer: ________________ cm

3. **Given:** $5(x - 2) = 3x + 8$. **Prove:** $x = 9$.
Complete both columns.

Statements	**Reasons**
1. $5(x - 2) = 3x + 8$	1. Given
2. $5x - 10 = 3x + 8$	2.
3.	3. Subtraction Property of Equality
4. $2x = 18$	4.
5. $x = 9$	5.

Answer: ________________

4. M is the midpoint of $\overline{AB}$, with $AM = 4x - 1$ and $MB = 2x + 7$ centimetres.

(a) Write the two-column proof that finds x. Start from the definition of midpoint and give a reason for every line.

(b) $x =$ ________ Then find AM.

Answer: ________________ cm

5. A, B, C and D lie on one line in that order, with $AB = 3x$, $BC = x + 2$, $CD = 2x - 1$, and $AD = 37$ centimetres.

(a) Name the postulate that lets you add the three short segments to get AD, then write and solve the equation: $x =$ ________

(b) Find the length BC.

Answer: ________________ cm

6. $\overline{AB} \cong \overline{CD}$, with $AB = 3x + 4$ and $CD = 5x - 12$ centimetres.

(a) Complete the proof below, filling every blank cell.

Statements	Reasons
1. $\overline{AB} \cong \overline{CD}$	1. Given
2. $3x + 4 = 5x - 12$	2.
3.	3. Subtraction Property of Equality
4. $16 = 2x$	4.
5. $x = 8$	5.

(b) Find the common length AB.

Answer: ________________ cm

7. Find the mistake. Jordan was asked to prove that $2(x + 3) = 4x - 10$ forces $x = 8$, and turned in this proof.

Statements	Reasons
1. $2(x + 3) = 4x - 10$	1. Given
2. $2x + 6 = 4x - 10$	2. Distributive Property
3. $6 = 4x - 10$	3. Subtraction Property of Equality
4. $16 = 4x$	4. Addition Property of Equality
5. $x = 4$	5. Division Property of Equality

Name the first line that is wrong, say exactly what the property requires, then write the corrected proof and give the true value of x.

Answer: ________________

8. Synthesis challenge. A, B, C and D lie on one line in that order, and $\overline{AB} \cong \overline{CD}$.

(a) Write a two-column proof that $\overline{AC} \cong \overline{BD}$. Use the Segment Addition Postulate on each of AC and BD, and say which property lets you add the same length to both sides.

(b) Now use numbers: $AB = 2x - 3$ and $CD = x + 5$ centimetres. Find $x =$ ________ and then the common length $AB =$ ________ centimetres. If $BC = 6$ centimetres, find AC.

Answer: ________________ cm